



Lecture 10

10.1 Laplace Transform for Solving ODEs

10.2 Further Properties of the Laplace Transforms

10.3 Convolutions and Laplace Transforms

10.1 Laplace Transform for Solving ODEs

常微分方程式を解くためのラプラス変換

10.1 Laplace Transform for Solving ODEs

Let us now discuss how the Laplace Transform method solves ODEs and initial value problems.

We consider an initial value problem.

$$y'' + ay' + by = r(t), \quad y(0) = K_0, \quad y'(0) = K_1$$

constants



where a and b are constant. Here $r(t)$ is the given input applied to the system and $y(t)$ is the output (*response to the input*) to be obtained.

10.1 Laplace Transform for Solving ODEs

Theorem 7.4 Laplace Transform of Derivatives

(i) Suppose that f is continuous on $[0, \infty)$ and of exponential order as in (7.3).

Suppose further that f' is piecewise continuous on $[0, \infty)$ and of exponential order. Then

$$\mathbf{l}(f') = s\mathbf{l}(f) - f(0) \quad (7.8)$$

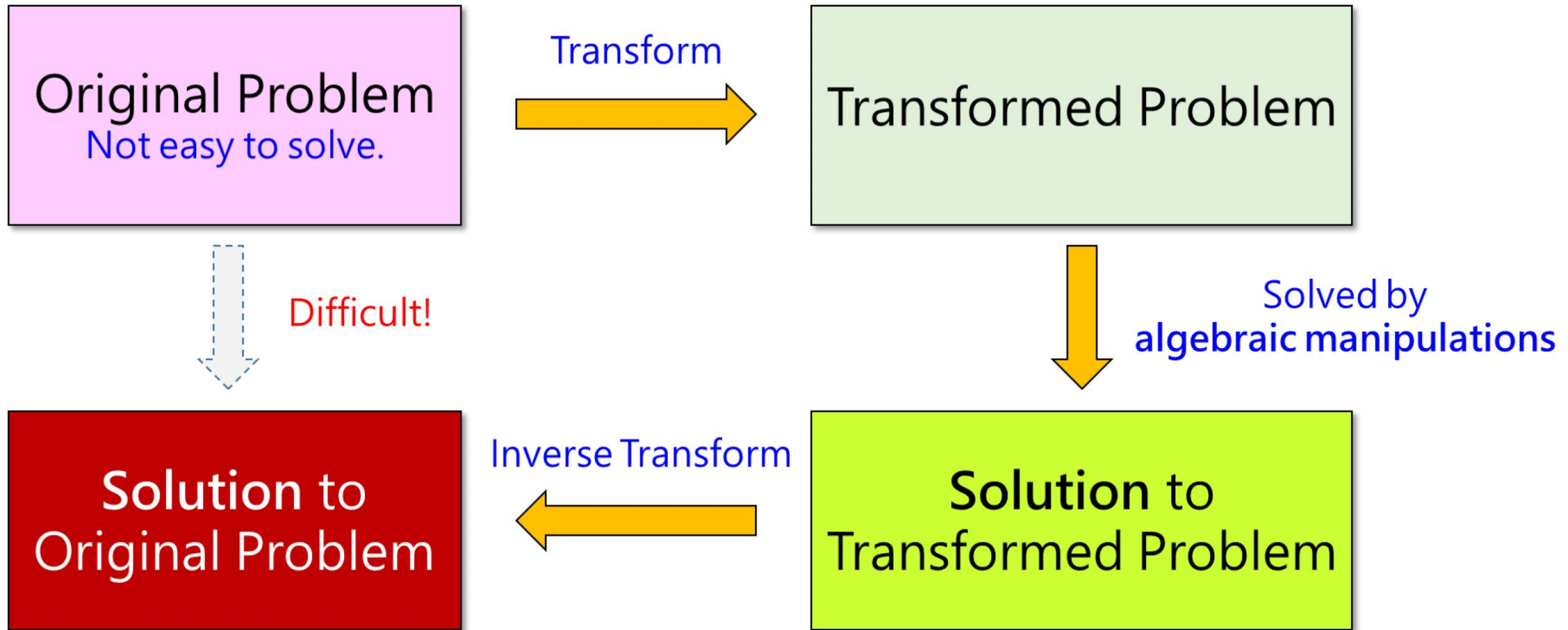
(ii) More generally, if $f, f', \dots, f^{(n-1)}$ are continuous on $[0, \infty)$ and of exponential order as in (7.3), and $f^{(n)}$ is piecewise continuous on $[0, \infty)$ and of exponential order, then

$$\mathbf{l}(f^{(n)}) = s^n \mathbf{l}(f) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0) \quad (7.9)$$

When $n = 2$, (7.9) gives

$$\mathbf{l}(f'') = s^2 \mathbf{l}(f) - sf(0) - f'(0) \quad (7.10)$$

10.1 Laplace Transform for Solving ODEs



10.1 Laplace Transform for Solving ODEs

$$y'' + ay' + by = r(t), \quad y(0) = K_0, \quad y'(0) = K_1$$


We solve the problem by doing **three steps**:

Step 1. Setting up the subsidiary equation, namely take the Laplace Transform for both sides of the differential equation and use the Theorem 7.4

$$[s^2Y - sy(0) - y'(0)] + a[sY - y(0)] + bY = R \quad (10.1)$$

Y is the Laplace Transform of y
 R is the Laplace Transform of r

$$\Rightarrow (s^2 + as + b)Y - (s + a)y(0) - y'(0) = R$$

$$\Rightarrow (s^2 + as + b)Y = (s + a)y(0) + y'(0) + R$$

Step 2. Find the solution of the subsidiary equation by algebraic manipulation

$$Y = \frac{(s + a)y(0) + y'(0) + R}{s^2 + as + b}$$

Step 3. Find the Inversion of Y to obtain $y = \mathcal{L}^{-1}(Y)$

We may reduce the original function by partial fractions to a sum of terms whose inverse can be found from the Table of the Laplace Transforms.

10.1 Laplace Transform for Solving ODEs

Example 10.1 A second order ordinary differential equation

$$\text{Solve } y'' + y = 2, y(0) = 0, y'(0) = 1$$

Solution

Step 1. Setting up the subsidiary equation, namely take the Laplace Transform for both sides of the differential equation and use the Theorem 7.4

$$\begin{aligned} [s^2 Y - sy(0) - y'(0)] + Y &= \mathcal{L}(2) \\ [s^2 Y - s \cdot 0 - 1] + Y &= \frac{2}{s} \\ (s^2 + 1)Y &= \frac{2}{s} + 1 \end{aligned} \quad \left| \quad \mathcal{L}(1) = \frac{1}{s} \quad s > 0 \quad \Rightarrow \quad \mathcal{L}(2) = 2 \cdot \frac{1}{s} = \frac{2}{s} \quad s > 0 \right.$$

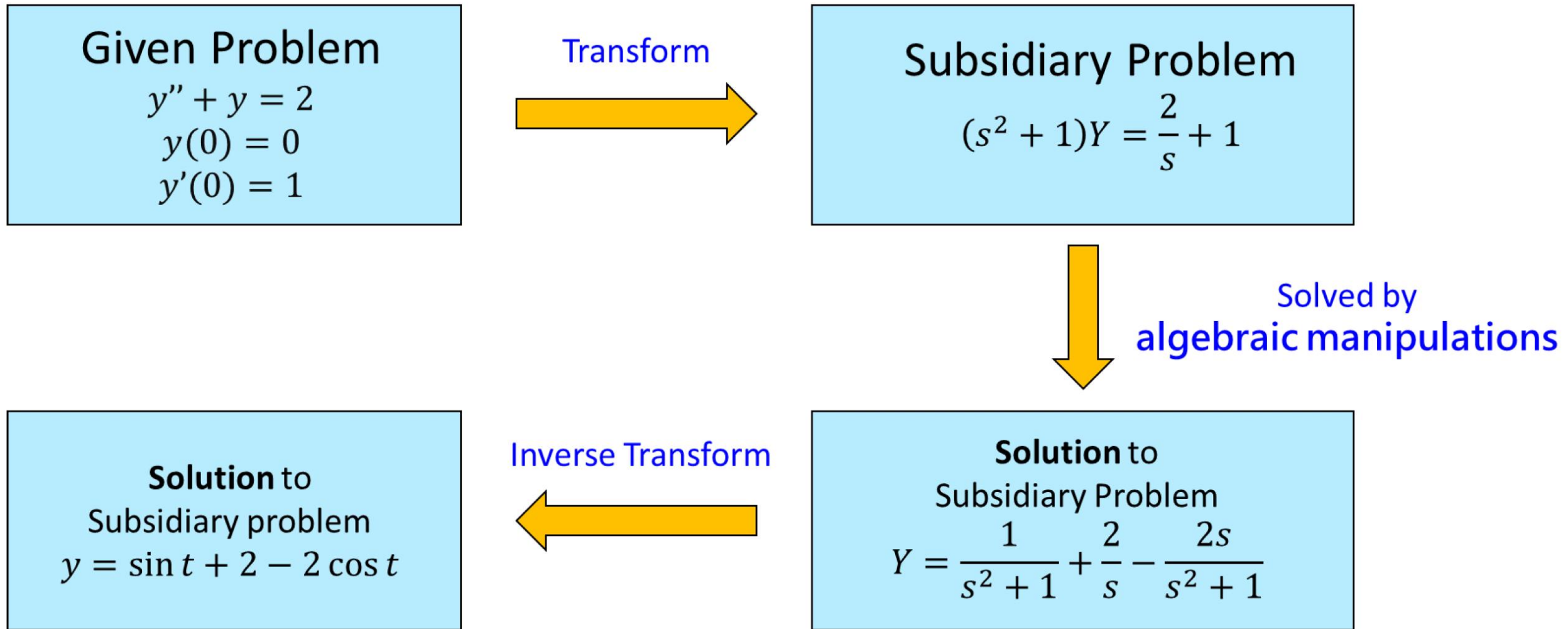
Step 2. Find the solution of the subsidiary equation by algebra

$$Y = \frac{\frac{2}{s} + 1}{s^2 + 1} = \frac{1}{s^2 + 1} + \frac{2}{s(s^2 + 1)} = \frac{1}{s^2 + 1} + \frac{2}{s} - \frac{2s}{s^2 + 1}$$

Step 3. Find the Inversion of Y to obtain

$$y = \mathcal{L}^{-1}(Y) = \mathcal{L}^{-1}\left(\frac{1}{s^2 + 1}\right) + \mathcal{L}^{-1}\left(\frac{2}{s}\right) - \mathcal{L}^{-1}\left(\frac{2s}{s^2 + 1}\right) = \sin t + 2 - 2 \cos t$$

10.1 Laplace Transform for Solving ODEs



10.1 Laplace Transform for Solving ODEs

Example 10.2 Shifting the time variable

Solve $y'' + 2y' + y = t, y(1) = 0, y'(1) = 0$

Solution

Making the change of variables $t = \tau + 1$, then $\tau = t - 1$

$$y'' + 2y' + y = \tau + 1, y(0) = 0, y'(0) = 0$$

Step 1.

$$[s^2 Y - sy(0) - y'(0)] + 2[sY - y(0)] + Y = \mathcal{L}(\tau) + \mathcal{L}(1)$$

$$(s^2 + 2s + 1)Y = \frac{1}{s^2} + \frac{1}{s}$$

Step 2.

$$Y = \frac{\frac{1}{s^2} + \frac{1}{s}}{s^2 + 2s + 1} = \frac{\frac{s+1}{s^2}}{(s+1)^2} = \frac{1}{s^2(s+1)} = \frac{1}{s+1} - \frac{1}{s} + \frac{1}{s^2}$$

Step 3.

$$y = \mathcal{L}^{-1}(Y) = \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) - \mathcal{L}^{-1}\left(\frac{1}{s}\right) + \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) = e^{-\tau} - 1 + \tau$$

Making the change of variables $\tau = t - 1$

$$y = e^{1-t} + t - 2$$

By Table of Laplace Transform

$$\mathcal{L}(\tau) = \frac{1}{s^2} \quad \mathcal{L}(1) = \frac{1}{s}$$

10.1 Laplace Transform for Solving ODEs

Advantages of our method

1. Solving a nonhomogeneous ODE does not require first solving the homogeneous ODE.
2. Initial values are automatically taken care of.
3. Complicated inputs $r(t)$ (right sides of linear ODEs) can be handled very efficiently.

*Homogeneous and nonhomogeneous equations

Recall that a linear differential equation

$$f_n(x)y^{(n)} + \cdots + f_1(x)y' + f_0(x)y = g(x)$$

was called **homogeneous** if $g(x) = 0$, and **nonhomogeneous** otherwise.

10.2 Further Properties of the Laplace Transforms

ラプラス変換のその他のプロパティ

10.2 Further Properties of the Laplace Transforms

We introduce an auxiliary function:

the *unit step function* or *Heaviside function*

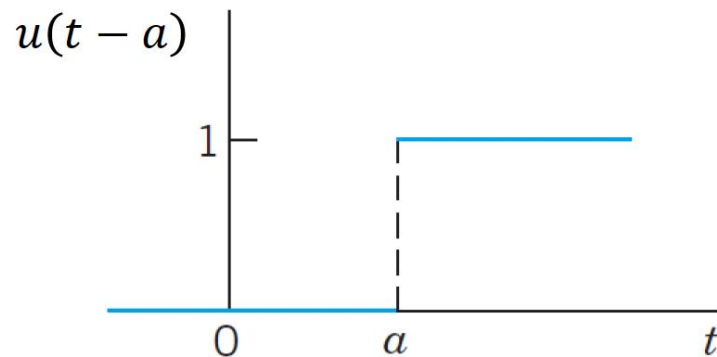
These functions are suitable for solving ODEs with complicated right sides of considerable engineering interest, such as single waves, inputs (driving forces) that are discontinuous or act for some time only, periodic inputs more general than just cosine and sine, or impulsive forces acting for an instant.

10.2 Further Properties of the Laplace Transforms

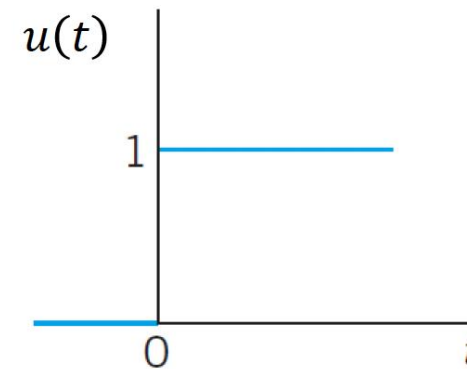
Unit Step Function (Heaviside Function)

The **unit step function** or **Heaviside function** $u(t - a)$ is 0 for $t < a$, has a jump of size 1 at $t = a$ (where we can leave it undefined), and is 1 for in a formula

$$u(t - a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases} \quad a \geq 0 \quad (10.2)$$



(a) Unit step function $u(t - a)$



(b) Unit step function $u(t)$, when $a = 0$

Figure 10.1 Unit step function

10.2 Further Properties of the Laplace Transforms

Given a function $f(t)$, consider the product $u(t - a)f(t - a)$.

Written explicitly, we have

$$u(t - a)f(t - a) = \begin{cases} 0 & \text{if } t < a \\ f(t - a) & \text{if } t \geq a \end{cases} \quad (10.3)$$

Thus, if $f(t)$ represents, say a signal, then $u(t - a)f(t - a)$ represents the same signal, but delayed by a units of time.

10.2 Further Properties of the Laplace Transforms

Theorem 10.1 Shifting on The t -Axis

If a is a positive real number, then

$$\mathcal{L}(u(t-a)f(t-a))(s) = e^{-as}L(s) \quad (10.4)$$

where $L(s) = \mathcal{L}(f(t))(s)$.

Proof

$$\begin{aligned} \mathcal{L}(u(t-a)f(t-a))(s) &= \int_a^\infty f(t-a)e^{-st}dt \\ &= \int_0^\infty f(T)e^{-s(T+a)}dT \quad (\text{where } t-a = T, dt = dT) \\ &= e^{-as} \int_0^\infty f(T)e^{-sT}dT \\ &= e^{-as}L(s) \end{aligned}$$

10.2 Further Properties of the Laplace Transforms

The unit step function is a typical “**engineering function**” made to measure for engineering applications, which often involve functions (mechanical or electrical driving forces) that are either “off” or “on.”

10.2 Further Properties of the Laplace Transforms

Example 10.3 Laplace Transforms involving unit step functions

(a) Evaluate $\mathcal{L}(u(t - a))$

(b) Evaluate $\mathcal{L}(f(t))$ where

$$f(t) = \begin{cases} 2 & \text{if } 1 \leq t < 4 \\ 0 & \text{otherwise} \end{cases}$$

10.2 Further Properties of the Laplace Transforms

Solutions

(a) The transform of follows directly from the defining integral in

$$\mathcal{L}(u(t-a)) = \int_0^{\infty} u(t-a)e^{-st} dt = \int_a^{\infty} 1 \cdot e^{-st} dt = -\frac{e^{-st}}{s} \Big|_a^{\infty}$$

here the integration begins at $t = a (\geq 0)$ because $u(t-a)$ is 0 for $t < a$. Hence

by Theorem 10.1 $\mathcal{L}(u(t-a)) = \mathcal{L}(u(t-a) \cdot 1) = e^{-as} \frac{1}{s} \quad s > 0$

(b) From the Figure 10.2, we can verify that

$$f(t) = 2(u(t-1) - u(t-4))$$

Then $\mathcal{L}(f(t)) = 2(\mathcal{L}(u(t-1) \cdot 1) - \mathcal{L}(u(t-4) \cdot 1))$

(By Theorem 10.1) $= 2 \left(e^{-s} \frac{1}{s} - e^{-4s} \frac{1}{s} \right) \quad s > 0$

$$= \frac{2}{s} (e^{-s} - e^{-4s}) \quad s > 0$$

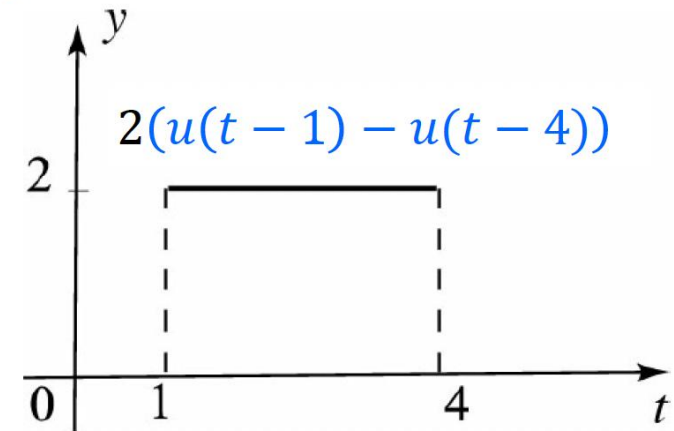


Figure 10.2 Example 10.3(b)

10.2 Further Properties of the Laplace Transforms

Example 10.4

Evaluate the Laplace transform of the ramp function shown in Figure 10.3.

Solution

For $t > 1$, we have $f(t) = u(t - 1)$, and for $0 < t < 1$, we have $f(t) = t(u(t) - u(t - 1))$. Verify that, by combining these two formulas, we have

$$f(t) = t(u(t) - u(t - 1)) + u(t - 1)$$

Rewrite $f(t)$ as follows:

$$f(t) = -(t - 1)u(t - 1) + tu(t) = -(t - 1)u(t - 1) + t$$

Recall that $\mathcal{L}(t) = \frac{1}{s^2}$. And by Theorem 10.1,

$$\mathcal{L}((t - 1)u(t - 1)) = e^{-s}\mathcal{L}(t) = e^{-s}\frac{1}{s^2}$$

Hence

$$\mathcal{L}(f(t)) = \mathcal{L}(-(t - 1)u(t - 1) + t) = -\frac{e^{-s}}{s^2} + \frac{1}{s^2}$$

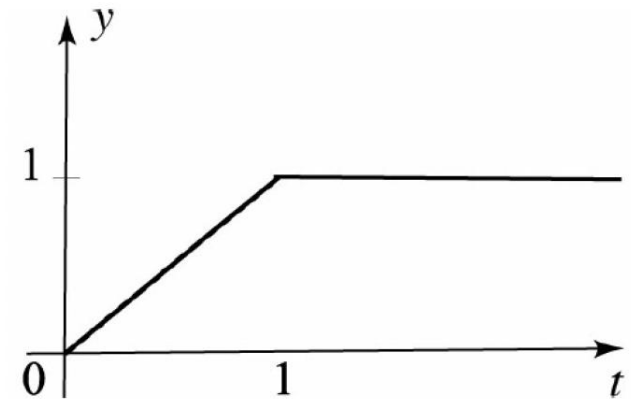


Figure 10.3 A ramp function

10.2 Further Properties of the Laplace Transforms

Example 10.5 A nonhomogeneous differential equation

Solve $y'' + y = f(t)$, $y(0) = 0$, $y'(0) = 0$, where $f(t)$ is as in Example 10.4.

$$f(t) = -(t-1)u(t-1) + tu(t) = -(t-1)u(t-1) + t$$

Solution

Step 1. From (10.1) and Example 10.4

$$(s^2Y - 0 - 0) + Y = -\frac{e^{-s}}{s^2} + \frac{1}{s^2}$$

$$(s^2 + 1)Y = \frac{1 - e^{-s}}{s^2}$$

Step 2.

$$Y = \frac{1 - e^{-s}}{s^2(s^2 + 1)} = (1 - e^{-s}) \left(\frac{1}{s^2} - \frac{1}{s^2 + 1} \right) = \frac{1}{s^2} - \frac{1}{s^2 + 1} - e^{-s} \left(\frac{1}{s^2} - \frac{1}{s^2 + 1} \right)$$

Step 3. Using Inverse Laplace Transform and Theorem 10.1, we have

$$\begin{aligned} y = \mathcal{L}^{-1}(Y) &= \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) - \mathcal{L}^{-1}\left(\frac{1}{s^2 + 1}\right) - \mathcal{L}^{-1}\left(e^{-s}\left(\frac{1}{s^2} - \frac{1}{s^2 + 1}\right)\right) \\ &= t - \sin t + u(t-1)[(t-1) - \sin(t-1)] \end{aligned}$$

10.3 Convolutions and Laplace Transforms

畳み込みとラプラス変換

10.3 Convolutions and Laplace Transforms

In this lecture, given two functions f and g , defined for all $t > 0$, we define their **convolution** $(f * g)(t)$ by

$$(f * g)(t) = \int_0^t f(t - \tau)g(\tau)d\tau \quad \text{for all } t \geq 0 \quad (10.5)$$

10.2 Further Properties of the Laplace Transforms

Theorem 10.2 Laplace Transforms of Convolutions

Suppose that f and g are piecewise continuous and of exponential order

$$\mathcal{L}(f * g) = \mathcal{L}(f)\mathcal{L}(g) \quad (10.6)$$

Proof

If we extend the functions f and g to be zero for $t < 0$, then the integral in (10.5) is same as

$$\int_0^{\infty} f(t - \tau)g(\tau) d\tau$$

Thus, throughout this proof, we assume that f and g are extended to the whole line with $f(t) = 0$ and $g(t) = 0$, For all $t < 0$. We have

$$\begin{aligned} \mathcal{L}(f * g)(s) &= \int_0^{\infty} \left[\int_0^{\infty} f(t - \tau)g(\tau) d\tau \right] e^{-st} dt \\ &= \int_0^{\infty} \int_0^{\infty} f(t - \tau)e^{-st} dt g(\tau) d\tau \quad (\text{interchange order of integration}) \\ &= \int_0^{\infty} \int_0^{\infty} f(u)e^{-s(u+\tau)} du g(\tau) d\tau \quad (u = t - \tau, du = dt) \\ &= \int_0^{\infty} \int_0^{\infty} f(u)e^{-su} du g(\tau)e^{-s\tau} d\tau = \mathcal{L}(f)\mathcal{L}(g) \end{aligned}$$

10.2 Further Properties of the Laplace Transforms

Example 10.6 Laplace Transforms involving convolutions

Evaluate $\mathcal{L}\left(\int_0^t (t - \tau) \sin(\tau) d\tau\right)$

Solution

From Theorem 10.2, we have

$$\mathcal{L}\left(\int_0^t \underbrace{(t - \tau) \sin(\tau)}_{f(t - \tau)g(\tau)} d\tau\right) = \mathcal{L}(t * \sin t) = \mathcal{L}(t)\mathcal{L}(\sin t) = \frac{1}{s^2} \frac{1}{s^2 + 1}$$


$$f(t - \tau)g(\tau)$$


$$f(t) = t \text{ and } g(\textcolor{red}{t}) = \sin t$$

10.2 Further Properties of the Laplace Transforms

Example 10.7 Solving differential equations with convolutions

Express the solution of the initial value problem.

$$y'' - 2y' + 5y = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

as a convolution.

Solution

Step 1. From (10.1)

$$(s^2 Y - 0 - 0) - 2(sY - 0) + 5Y = L(s)$$

$$(s^2 - 2s + 5)Y = L(s)$$

Step 2.

$$Y = \frac{L(s)}{s^2 - 2s + 5} = L(s) \frac{1}{(s-1)^2 + 2^2} = L(s) \left[\frac{1}{2} \frac{2}{(s-1)^2 + 2^2} \right]$$

Step 3. Find the Inverse Laplace Transform in the table and using Theorem 10.2, we have

$$\begin{aligned} y = \mathcal{L}^{-1}(Y) &= \mathcal{L}^{-1} \left(L(s) \left[\frac{1}{2} \frac{2}{(s-1)^2 + 2^2} \right] \right) = f(t) * \frac{1}{2} e^t \sin 2t \\ &= \frac{1}{2} \int_0^t e^{t-\tau} \sin 2(t-\tau) f(\tau) d\tau \end{aligned}$$

Review for Lecture 10

- Laplace Transform for Solving ODEs
- Laplace Transform for The Function Shifting on The t Axis
- Convolution and Laplace Transforms

Assignment

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

Reading materials: Lecture 10 Slides

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004